

Manual Sketching: Why is it Still Relevant??Revisited

Miriam Sturdee
University of St Andrews
St Andrews, United Kingdom
ms535@st-andrews.ac.uk

Makayla Lewis
Kingston University London
London, United Kingdom
m.m.lewis@kingston.ac.uk

Denise Lengyel
Uppsala University
Uppsala, Sweden
denise.lengyel@im.uu.se

Abstract

In 2009, Goldschmidt explored the role of sketching within the emerging landscape of digital drawing tools and Computer Aided Design (CAD), concluding that manual sketching remained fundamentally important. In 2025, this question demands renewed attention: In an era of smartphones, the Internet of Things (IoT), and a Generative Artificial Intelligence (GAI) goldrush, images can be created instantly from text or speech. So is manual sketching still relevant in the face of automated creative output? Does it retain an irreplaceable quality? Or is a new hybridity emerging? Is it naïve—or even irresponsible—to ignore the digital transformation shaping creative practice today? And what costs come with this transformation? Based on preliminary results from two recent case studies, we reflect on this, examining the evolving role of sketching for practitioners, researchers, and students, illuminating how we choose to navigate the complex intersection of human creativity and machine generation.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

sketching, GAI, UX

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1 Introduction

Human beings like to leave their mark on the world, and during the act of shaping the world around them, they leave traces, marks and imprints, by conscious or unconscious action. Sketching is one such conscious action, put simply, transmitting human thought to another medium, via making lines and shapes. Sketching today is a far cry from drawing lines in the dirt, or daubing the walls of caves to tell stories: One might pick up a stylus rather than a pencil, create virtual layers on a tablet, or even sketch imagined world live on stage using a VR headset¹. Traditionally, sketching

is defined as the act of creating quick, informal visual representations to explore, communicate, and develop ideas, typically using lines, shapes, and minimal detail [13, 15, 16, 18]. In HCI (Human Computer Interaction), sketching can take on a range of diverse roles, such as rapid prototyping, design, and even AI-augmented creativity [13, 15, 16, 18], with pen and paper also taking on a new-found 'digital' role within the context of HCI, with simple sketches being processed through algorithms and converted into a 2D or 3D digital representation, or even being applied as a form of input 'as is' [18, 20]. Despite the many advances in the technology of mark making, there still remains a fundamental connection between the maker and the medium: movement and action creates traces and imagery, and thus a direct translation of the will of the maker. So what happens when you digitise and thus remove one (or more) of the steps? How will 'traditional' and new technological sketching tools, and their joint use, continue to change the role, process and outcome of design? Will this become a hybrid practice governed by "when and what for", "freedom versus efficiency" and the respective tool constraints like [19] observed for drawing and design after the technological push introduced by Computer Aided Drawing (CAD)? In search for an answer, let us have a renewed look at Goldschmidt's perspective on said technological push and its influence on the role of manual sketching.

2 Revisiting 'Manual Sketching: Why is it Still Relevant?'

When Goldschmidt asked this question in 2017 [10], CAD had already been revolutionising how architects and engineers approached design and visualisation. The tablet was replacing the pen and paper as an approach to illustrative means, and powerful image editing tools allowed the manipulation of found imagery. The concern was, that sketching was becoming 'devalued' despite it having something fundamental to the early design process that was being missed when individuals moved straight to the precise, angled and measured lines of CAD and its fellow advanced visualisation programs. Until the digital revolution, little had changed for ideation sketching in over 500 years, but could it now be replaced? The paperless movement alarmed traditionalists in 2012 (*Vital Signs: Is Drawing Dead?* [14]) and now we are, in turn, alarmed by the 'humanless' movement.

In 2012 the debate oscillated between those who believed sketching and drawing on paper was indispensable and insurmountable, and those who believed that it could –and should– be replaced by the newly available digital tools. The difference between then and now is that digital drawing still requires human cognition – the 'mind-eye-hand' combination which Goldschmidt believes essential: computation interrupts this fundamental process. Sketching is not merely images in the mind, but a visual display that helps 'induce' an image, a form of pictorial reasoning that gradually leads

¹www.annadreambrush.com/



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to design [8]. However, computation is tempting, especially for a technology savvy audience who see Generative Artificial Intelligence (GAI) as ‘leading the way’ in the design industry [10]. In Goldschmidt’s discourse, young people saw the power of the computer as tempting as its use could take the lead in design – reducing pressure on the person. Goldschmidt noted that students believed that current practice in industry was working with digitally created and projected ideas, and railed against the teacher’s intervention with pen and paper. I (author 1) note that one of my own students said there was far too much focus on sketching in our industry facing User Experience (UX) course, and that the time would be better spent learning the multitude of digital tools and indulging in vibe coding to produce fully formed solutions – missing the point in hand. Sketching allows student and teacher not only to ‘achieve a shared mental model’ [10] but also to ‘extend and broaden arguments’ between student or teacher, or even converse [6]. Sketching helps a shared vision be shaped by all involved, and allows input from multiple parties in quick succession.

Although information can be ‘read’ from other images, such as those found during internet search and displayed within a mood-board, or edited to reflect the thoughts of the designer, it does not compare to the sandbox of sketching where we might play and learn until we find the answer [8]. The copy-paste attitude (do we now have a prompt-output attitude?) also means that the underlying reason, the ‘why’ of image generation is lost. ‘Found’ imagery might also exclude things that cannot be searched for or anticipated – similar to how we might rely on only that which we can prompt – without backtalk, how might we innovate beyond what we can articulate? Found imagery also has its own issues in copyright, despite editing, re-drawing, changing context, there remains some forms of ownership, and regulation has adjusted to take into account the lawfulness of this ‘borrowing’. However, how does one ascribe authorship to that which is generated by AI [3]? Especially when it has repeatedly been found to ‘borrow’ almost exactly from working artists? There are also fundamentally advantageous qualities in self-generated imagery – sketching is a form of visual thinking, or further, a way of having a useful argument with oneself. Returning to Goldschmidt’s earlier work, this back-talk of self-generated sketches is inherent in their ongoing value – mistakes or rough lines can suggest multiple interpretations and inspire novelty or strangeness [9]. To err is human, not machine. It is hard to argue –or even play– with the ‘black box’ of GAI, despite ongoing inspirational work in Explainable AI [5].

Design requires visual thinking: can we ‘think’ visually if we do not create the imagery? Manual sketching is especially important for early ideation. Sketching can also interact with generated or found imagery, and this should be encouraged in cases where GAI has created that image – this to-and-fro can be beneficial, if we see the AI as a tool to assist the process rather than replace it, part of a conversation [25]. Sketching has tolerances – it revels in the incomplete, the rough and imprecise. This ambiguity leads to surprises. Where there is no room for mistakes, there can be no interrogation, merely a production line of imagery which may or may not fulfill the designers’ desires. The computer cannot be denied in its power, however to enable us to search for design ideas, even ideas than a human can generate, or complex forms which we lack the skills to visualise and build – but sketches can still work

in tandem with parametric software to design complexity (even if sometimes it goes wrong, e.g. Goldschmidt gives the example of Gehry’s Stata Center building ²) providing a layered approach to a solution. The same might be true of images generated by AI, which can serve as informative or inspirational [17] but do not necessarily replace the artist. The human-in-the-loop makes the choice to be replaced. Goldschmidt did not abhor the advances in technology that threatened to displace sketching, but instead stated that both have a place in the process, and this remains true within our own new digital revolution.

3 Exploring Assumptions

This late-breaking research preliminarily explores three modes of integrating GAI into sketching in UX and game design education and research: post-ideation GAI utilisation, staged GAI engagement, and minimal or post-ideation GAI use. Early analysis revealed distinct pedagogical frameworks, modes of student engagement, and sketching outcomes associated with the timing and role of GAI within the UX and game design ideation process. These preliminary findings, derived from Case Studies 1 and 2, guided the design of a third case study focused on Early GAI led Ideation, accompanied by an expanded data-collection program to enable a more comprehensive comparative analysis. This iterative research design fortifies the emerging analytical framework that addresses the question “Manual Sketching: Why Is It Still Relevant – Revisited” and contributes to an evolving discourse on the role of sketching and GAI in contemporary design pedagogy.

3.1 Who does it better? – Case Study 1

A student researcher was tasked with carrying out a pilot study and investigation asking ‘Why is Manual Sketching Still Relevant?’ to lay the ground for a further empirical study in comparative sketching between humans and GAI. Four students were recruited from a preexisting industry-facing class in which teams were presented with real design problems, and required to go through an abridged data elicitation and UX design process in response to a client design brief. The participants were asked to share their early ideation, storyboarding, or concept sketches. The same brief was then used to generate the same type of sketch using DALL-E ³ (at the time of the study, the newest versions of ChatGPT ⁴ or CoPilot ⁵ were not available). The participants were also asked to take part in a short interview to explain their process and thoughts about sketching and AI. This was followed up by a survey in which a new set of participants were recruited from the general student pool (n=27) and asked to judge both the human generated and AI generated sketches on a number of parameters: Clarity, Function, Visual Appeal, and Creativity. Three of the four participants took part in the interview sessions.

As a cohort, participants stated that while DALL-E outputs were occasionally visually appealing, they lacked clarity, intentional structure, and meaningful communication. In contrast, although they varied in quality (due to experience and/or confidence of the

²www.nytimes.com/2007/11/07/us/07mit.html

³Note that this url is the new version – the one we used is discontinued www.openai.com/index/dall-e-3/

⁴www.chatgpt.com

⁵www.copilot.microsoft.com

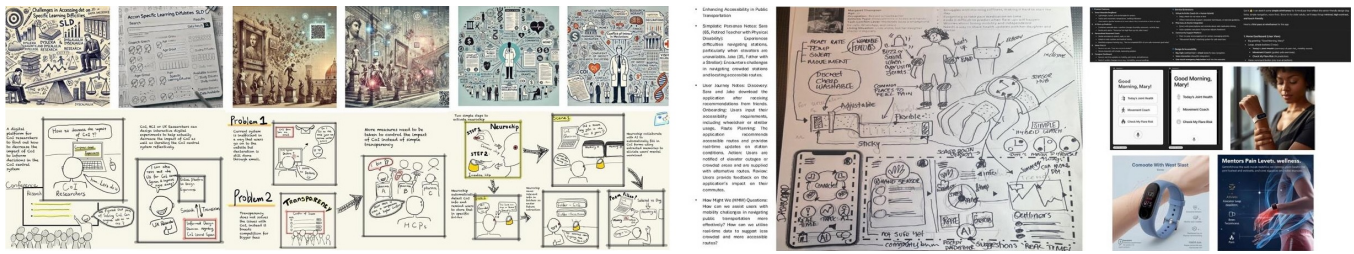


Figure 1: Left Collection: Examples of GAI responses to a client brief (top) compared with human response and visual narrative (bottom). Right Collection: Examples of (left) a workshop design challenge; (middle) an in-class demonstration of self-ideation; and (right, top to bottom) an in-class demonstration of prompts and GAI output using ChatGPT, a GAI sketch using ChatGPT, and a GAI visualisation created with Canva AI.

individuals) were seen to reflect a deeper understanding of the design brief, client needs, and purpose. The findings for the survey indicated that while AI-generated sketches offered fine detail, visual appeal and speed, they often lack clarity of structure, intended ambiguity and openness, and lacked the iterative potential of manual sketches. Further, the human-made sketches were consistently preferred by participants across three design briefs, with reference to their logical narratives, expressive and meaningful annotations, and because they were more closely aligned with the clients' briefs and end user needs. The AI generated 'sketches' were often described as visually confused or lacking semantic meaning, despite the appearance of finality and finesse. Overall, this work suggests a continued appeal and value of sketching for prototyping and ideation, whilst pointing toward AI as a means to create stylised and conceptual images: Sketching is a way of thinking and developing, and has cognitive benefits that appear to outweigh style and ease of creation.

3.2 Scratching the GAI Itch – Case Study 2

GAI tools, such as ChatGPT, Copilot, and Canva AI ⁶, have gained increasing prominence and, in certain instances, have been legitimately integrated into higher design education, influencing students' exploration of early concepts [4, 11, 22]. While GAI can accelerate idea generation, it also carries the risk of promoting superficial outcomes if introduced without critical framing. In this context, postgraduate students (n=140) enrolled in a ten-week, user experience design- and game design-focused, design thinking-oriented module in the United Kingdom demonstrated a strong inclination to employ GAI at every stage of the creative process (empathise, define, ideate, prototype, and test). This tendency, characterised as the "GAI itch" manifested as repeated requests to legitimise GAI use across all design, practical, and assessment work, driven by the need to incorporate GAI into every aspect of their work, partly due to the constant discussions of GAI's innovation and potential to support human capabilities, and to an equal extent, "impatience" in the design process, aiming to expedite tasks, reduce effort, and initiate high-fidelity prototyping [4].

The primary pedagogical challenge for higher education educators lies in balancing students' desire to experiment with GAI, their impatience, and their commitment to user-centeredness, reflective

sketching, and judicious decision-making. Unrestricted GAI use could bypass self-initiated ideation and user interpretation, echoing concerns raised in recent debates about the impact of GAI tools on creative work, e.g., [24]. Conversely, prohibiting GAI outright could lead to disengagement and disregard for its growing relevance in professional design practice [7, 17, 22]. Therefore, the challenge as an educator lay in designing an intervention that supported critical engagement with GAI without allowing it to dominate or supplant established ideation practices.

A structured ideation workshop was organised into four sequential phases: Self, Users, GAI, and Reflection. Before the workshop, students attended a preparatory lecture introducing ideation theory, sketching practices (with and without GAI, e.g. Figure 1), and comparative methods. They completed assigned readings addressing reflective design practice and the limitations of computational creativity (e.g., [13, 15, 17, 18]). The intervention was conducted during a three-hour workshop in which students worked in groups of four to five on a simplified but realistic design challenge. The first 30 minutes were dedicated to orientation and problem framing, during which the instructor introduced the design brief, clarified expectations, and reviewed the staged methodology. To promote exploratory reasoning and interpretive sketching, supporting materials—including personas, customer journey fragments, and "How Might We?" questions—were intentionally under-specified. During the subsequent 75 minutes, corresponding to the Self and Users phases, students completed self-driven, user-informed ideation activities through sketching and group discussion. They were not permitted to use GAI tools. During this period, the instructor circulated among groups, prompting students to externalise reasoning through sketches, encouraging consideration of multiple perspectives, and redirecting attention to user needs when discussions became solution-driven. No evaluative feedback was provided at this stage.

The GAI phase occupied the following 45 minutes. Before this phase began, the instructor provided a brief demonstration of example prompts and outlined expectations regarding the critical comparison of GAI-generated ideas. Students then used ChatGPT, Canva AI, and or Copilot to generate five candidate ideas for each "How Might We?" question, compared these outputs to select the most promising concepts, and expanded the chosen ideas through additional prompting focused on potential innovations, anticipated challenges, and solution development. Throughout this phase, the

⁶www.canva.com/ai-image-generator/

instructor intervened to discourage uncritical acceptance of GAI outputs, to prompt justification of selection decisions, and to remind groups to relate generated ideas back to earlier sketches and user insights. The last 30 minutes were allocated to individual and group reflection. The instructor provided guidance on reflective framing and criteria for critical evaluation. Each group nominated one student to present six reflection statements summarising insights from collaborative interactions and key experiences during the activity, limiting each statement to fewer than six sentences.

Early analysis of these reflections revealed that the structured approach fostered critical awareness of GAI's role in ideation, aligning with findings by [23]. Numerous students found GAI-generated ideas to be generic, repetitive, or lacking in user context, despite extensive prompt refinement. Several groups questioned whether the effort invested in prompting yielded a proportional creative return; their 'New, Useful and Feasible' (NUF) evaluations [12] indicated that their initial sketches were often more innovative and better aligned with the design brief. The enforced sequencing prompted students to articulate more precise distinctions between ideas arising from embodied sketching, user interpretation, and tool-driven generation. Rather than suppressing the "GAI itch", the workshop positioned GAI as a secondary and interrogative resource rather than a primary driver of ideation. These findings suggest that the timing and pedagogical framing of GAI utilisation influence not only students' evaluations of AI-generated ideas but also their perception of creative agency. Positioning GAI after self- and user-driven ideation re-framed the tool as a reflective resource rather than a primary source of ideas, supporting critical rather than substitutive use. This pattern aligns with accounts of embodied and external cognition in design, wherein sketching serves as a generative medium for sense-making rather than a representational endpoint. The selective benefits reported by less confident sketchers further extend prior research on technology acceptance. Consistent with [2] and the Technology Acceptance Model (TAM), acceptance was contingent upon perceived learner ability. However, whereas Chandrasekera observed higher acceptance among visual learners, the present case suggests that GAI may serve as a compensatory representational aid for students who self-identify as less visually proficient. In this context, GAI does not replace sketching but redistributes representational agency, enabling participation without displacing embodied ideation practices. Overall, this Self, Users, GAI, and Reflection workshop demonstrates how structured sequencing mediates the relationship between sketching and GAI, supporting the comparative claim that pedagogical timing facilitates discernment rather than defaulting to AI-generated ideas.

4 Invitation to Engage & Statement of Intent

Goldschmidt did not abhor the advances of technology, but saw them as co-existing together, and perhaps this is the form that sketching will take in the future, once we acclimatize to GAI's perpetuation of ephemeral and disconcertingly precise imagery. We could also be missing something however: GAI might only replace sketching if we *want* it to, and we *allow* it to. The two case studies illustrated here show two sides to perhaps what is a three sided debate: the first concludes that manual sketching is difficult to replace – the lack of intentional abstraction and tendency to

enable design fixation means human-led sketching is key [26], the second that it is a necessary part of modern design, but can be used in tandem [17]. The third side might be complete replacement of the human in the design and sketching process, or perhaps, remove the need for sketching entirely if 'serendipitous discovery' and the stimulation of visual output can be channeled via GAI [21].

This late-breaking research presents several implications for integrating GAI and sketching in design education. Firstly, instructional design should prioritise sequencing over accessibility: learning outcomes are more strongly shaped by the order in which tools are introduced than by their accessibility. Sketching should be retained as a primary sense-making activity, with GAI positioned as a secondary resource that supports rather than supplants authorship over emerging concepts. Pedagogical designs should explicitly encourage comparative reflection, rather than assuming that students will naturally evaluate GAI outputs critically. Framing GAI as an interrogative rather than an authoritative tool may support critical thinking, iteration, and justification, thereby reinforcing human agency in creative decision-making. These implications extend to instructor preparation and tool design.

Higher education providers should include instructor training that emphasises integration rather than substitution or exclusion, demonstrating how GAI can complement rather than replace sketch-based ideation. Instructors should also design explicit moments of reflection that highlight the strengths, limitations, and biases of both human and GAI contributions throughout the design process. Future GAI design tools should similarly prioritise transparency, comparison, and justification to support reflective creative practice (e.g., [1]). These initial explorations have led us to design an empirical study that directly compares human and GAI sketching using authentic design prompts. We pose the following questions: how do humans and GAI differ in the sketches they generate, how are these outputs interpreted, and when are such results perceived as meaningful? We now extend this inquiry to the CHI community: where do you position yourself between sketching and GAI? We invite you to join us at the poster session to explore this comparison. Please submit your sketches to the poster, share your prompting strategies, and contribute to the ongoing debate on how sketching and GAI should coexist in future design practice.

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